September 4, 2026

Prof. Roland Rathelot

Editor

The Economic Journal

Subject: Revision of "Boots in the Beat: Effects of Beat Policing on Victimization and Crime Perception"

Dear Prof. Rathelot,

Thank you for giving us the opportunity to revise our manuscript further. We are delighted that Referee 3 has signed off on the paper, and we carefully read your comments and the comments from Referees 1 and 2. We are very grateful for this detailed and constructive feedback, which helped us further improve the paper.

Below we provide a point-by-point response document with detailed answers to each comment. We also indicate how the manuscript has been revised in response.

Sincerely,

Andrés Barrios Fernández on behalf of the team

Editor's comments

Editor Comment 1

Dear Doctor Barrios-Fernandez,

Thank you for submitting your fascinating paper to The Economic Journal.

I have now heard back from the reviewers and read your revised paper. The news is rather good. R3 has signed off on the paper. R2 has a few comments, which are rather conditional-acceptance material. R1 is pleased with the revision but still has issues regarding the interpretation of your results and the overall framing of the paper. I agree with them. I think we have established that your paper makes a contribution to the literature; now is the time to acknowledge very precisely what can and cannot be said about the results. Please provide further evidence to support the current interpretation of the results if you can and if the evidence supports it (the reviewer has suggestions). Also, please revise the framing so that the results, interpretations, and framing are well aligned. At this stage, I would like you to be mindful of readers and start condensing your paper to maximise readability and impact. I think there is room to cut around 5 pages and to make the introduction (and the abstract) shorter, in particular (of course, do not cut anything you find crucial).

You will find detailed comments in R1's and R2's reports. As usual, I would like you to prepare a detailed, separate reply for each reviewer and revise the paper accordingly. Additionally, please prepare a separate note for me, outlining the changes made during the revision process.

Response to Editor Comment 1:

References

Referee 1 Suggestions, Comments and Questions

Thank you very much for your careful reading of the revised manuscript and for finding the causal evidence on the effects of *Plan Cuadrante* convincing. We have carefully considered your remaining comments on interpretation and framing and hope that we adequately addressed them in the revised manuscript and in the response document below.

Comment R1.1

1. What does Plan Cuadrante identify? The paper states in various places that permanent geographic assignment and greater local knowledge are the key mechanisms underlying the RF effects of the policy. I think this interpretation is stronger than what the evidence can establish.

Plan Cuadrante appears to have changed several features of frontline policing simultaneously. Assigning officers to smaller quadrants changes not only the officer-area relationships, but also the spatial organization of police presence. Depending on quadrant size, number of officers, and deployment practices, the reform may have effectively changed police density and patrol intensity, as well as response times. Each of these margins could plausibly affect crime, victimization, perceptions of safety, and citizen cooperation. Greater local knowledge is one compelling mechanism within this bundle, but it is not separately observed from these other operational changes.

I therefore think the paper should explicitly acknowledge that the results identify the effect of a broader organizational and deployment reform rather than the effect of local knowledge alone. A more defensible interpretation would be that Plan Cuadrante, combining permanent officer-to-quadrant allocation with an increase in police density, improved public safety and citizens' perceptions of the police. Even holding the total number of officers fixed, reorganizing officers across smaller territorial units may change the intensity and concentration of local police presence.

One potentially useful exercise would be to characterize police density across quadrants, for example in terms of officers per square kilometer (or per capita). The illustrative map in the paper suggests that quadrants vary substantially in geographic size (and the smaller quadrants are the ones with higher baseline crime rates). If the authors observe the number of officers assigned to each quadrant, it would therefore be useful to document the distribution of officers per quadrant, and examine whether treatment effects vary with this measure of effective local police density. Police density may itself be correlated with response times, patrol intensity, and other operational margins, so it would still not isolate a single mechanism. But it would help characterize an important dimension of what the reform changed.

Response to R1.1:

Dani: Podemos asignar los policias o los recursos a un cuadrante determinado? Dejame pensar.

Thank you very much for your comment and suggestion.

We agree that even if the total number of officers is held fixed, the reorganization of officers across smaller territorial units may change the intensity and concentration of local police presence, increasing it in some areas, and decreasing it in others. If the allocation of police concentration is now more efficient, the effect might be at least partly driven by this relocation. Unfortunately, to test this hypothesis, we would need to have data on the density of police officers in each quadrant both before and after the implementation of Plan Cuadrante. We previously requested this data, but unfortunately that was not granted, neither before nor after the introduction of Plan Cuadrante.

Thus, at this point, the best we could do is rephrase the interpretation of the results. In XXX, where it says....

OTROS EJERCICIOS: CUADRANTES CON Y SIN COMISARIA DE POLICIA O MAS ALEJADOS DE DONDE ESTA LA COMISARIA.

Comment R1.2

2. Resource changes. Figure 4 is most informative in the comparison between the bottom 40 percent of treated municipalities and control municipalities, which experienced similar resource growth. The interpretation should nevertheless be cautious. In particular, not even the bottom group or the control group experience zero resource growth. The exercise therefore does not isolate the effects of the geographic reorganization, absent those of changes in resources. Rather, it shows that the estimated gains are not confined to municipalities experiencing the largest resource expansions.

Relatedly, I think Table B3 is helpful. I would not dismiss these specifications simply because resource changes are themselves induced by treatment. Conditioning on policy-induced resource changes can still be informative about whether the reduced-form effect is concentrated in places experiencing large resource expansions.

Response to R1.2:

Thank you very much for your comment and suggestion. We....

Comment R1.3

Minor point: in Panel D in Table B3, the sample size falls substantially when the officer control is added. Why?

Response to R1.3:

Thank you for pointing this out. This is due to an inconsistency in how the resource-years sample was defined in Panel D of the previous version of the paper. Let us start by explaining what we used, and how that explains the larger drop in the sample size in Panel D.

Unlike the outcomes in Panels A–C, the outcome in Panel D is based on administrative records of reported crime, which are available for a longer period (2002–2016) than the ENUSC survey data used in Panels A–C. These administrative records also cover never-treated municipalities, which allows the Callaway and Sant'Anna estimator to use the full period. This is not possible with the ENUSC-based panels, which are limited to 2003–2012 (excluding 2004, for which no ENUSC survey was conducted): with one exception, every municipality in the ENUSC estimation sample is already treated by 2012, so there are no valid comparison units left after that year, and post-2012 survey waves cannot contribute to identification regardless of resource-data availability.

Turning to the specifics of Panel D in Table B3 as it appeared in the previous version of the paper: column (1) reports the main estimate using the full 2002–2016 period. Column (2), which includes no resource control, uses 2006–2016. Column (3) is restricted to 2006–2012 once the officer control is added, since officer data are only available for that window. The four years that are lost, 2013–2016, account for 1,045 of the 1,052 municipality-year observations that disappear between columns (2) and (3); the remaining 7 correspond to the single municipality for which the number of officers is not reported. The same logic explains why column (4) is smaller than column (2) in Panel D but larger in the other panels: column (4) keeps 2005, since patrol-vehicle data are available from 2005 to 2012, but it also loses 2013–2016, and in the ENUSC panels there is nothing after 2012 to lose.

In Panels A–C, outcomes are measured at the household level, and the estimation period is 2003–2012 in column (1), 2006–2012 in columns (2) and (3), and 2005–2012 in column (4). In each case, the estimation window is limited by the availability of the resource variable entering that column.

One important inconsistency in the previous version of the paper is that the resource-years sample was not defined consistently across panels. In Panels A–C, it corresponds to the years for which both resource data (officers and patrol vehicles) are available, 2006–2012. In Panel D, however, it was defined as starting in 2006, the first year for which both officers and patrol vehicles are available, but running to the last year of the administrative panel, 2016, even though no resource data are available after 2012. We have corrected this inconsistency in the revised version of the paper by defining the resource-years sample as 2006–2012 uniformly across all panels.

Table I below reports the corrected table, as it now appears in the revised version of the paper, with the resource-years sample defined as 2006–2012 in every panel:

For comparison, Table II reproduces the table exactly as it appeared in the previous version of the paper, where column (2) of Panel D was not bounded above and therefore ran to 2016:

Only Panel D of column (2) differs between the two tables. Restricting that column to 2006–2012 reduces the estimation sample from 2,963 to 1,918 municipality-year observations and changes the estimated ATT from −157.010 (standard error 56.224) to −190.446 (standard error 67.908). The estimate remains negative and statistically significant at the 1% level, and is closer to the baseline estimate in column (1). Every other cell in the table is unchanged, since columns (1), (3), (4) and (5) were already restricted to their own data-availability window.

Comment R1.4

3. Contribution and framing. I would simplify the framing of the paper's contribution. The findings provide evidence that the organization and territorial deployment of police resources matter for public safety (and for citizens' perceptions, which by itself is a quite novel result), even though the specific mechanisms generating these gains cannot be fully decomposed. I think this is an interesting and important contribution on its own, and it does not require attributing the effects specifically to local knowledge.

Response to R1.4:

Thanks for the suggestion. Following reviewer's recommendation, we have edited key parts of the paper. The updated sentences read as follows:

XX XX

Table I: Effects of *Plan Cuadrante* Controlling for Police Resources

	(1)	(2)	(3)	(4)	(5)
Panel A. Victimization					
ATT	-0.099***	-0.077***	-0.086**	-0.094***	-0.087*
	(0.013)	(0.013)	(0.043)	(0.028)	(0.053)
N	93,041	53,973	53,553	72,017	53,553
Panel B. Crime perception					
ATT	-0.149***	-0.140***	-0.112**	-0.082**	-0.111
	(0.025)	(0.032)	(0.057)	(0.033)	(0.106)
N	89,578	52,160	51,746	69,434	51,746
Panel C. Investment in security measures					
ATT	-0.077***	-0.066***	-0.109***	-0.081***	-0.168***
	(0.016)	(0.019)	(0.041)	(0.019)	(0.064)
N	93,041	53,973	53,553	72,017	53,553
Panel D. Reported crime (per 100k pop.)					
ATT	-208.024***	-190.446***	29.230	-261.439	-245.555
	(56.130)	(67.908)	(147.261)	(170.954)	(165.221)
N	4,231	1,918	1,911	2,248	1,911
Sample	Full	Resource yrs	Resource yrs	Resource yrs	Resource yrs
Police officers (per 100k pop.)			✓		✓
Patrol vehicles (per 100k pop.)				✓	✓

Note: ATT estimates of the effect of *Plan Cuadrante* on each outcome. Column (1) is the baseline (full estimation sample, no controls) and reproduces the headline estimate of the corresponding estimator. Column (2) restricts the estimation to 2006–2012, the window over which police resources are observed, without including any control variable. Columns (3)–(5) add time-varying controls for police resources per 100,000 inhabitants: officers only; patrol vehicles only; and officers and patrol vehicles jointly (entered as separate regressors). Columns (3) and (5) cover 2006–2012, the years for which the number of officers is observed; column (4) also includes 2005, the first year for which patrol vehicles are observed. Patrol vehicles are measured, per 100,000 inhabitants, as all patrol automobiles (radiopatrulla, auto patrullero, auto policial, automovil, station wagon, jeep and camioneta) plus motorcycles. Standard errors in parentheses; row N reports the estimation-sample size. Estimation follows **?**; standard errors clustered at the municipality level using wild bootstrapping procedures. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Comment R1.5

By contrast, I do not find the analogy in the introduction to the spatial organization of health and education services particularly convincing. The underlying economic problem is different. Health and education services are typically supplied at fixed locations, with citizens incurring costs to access them, and supply being constrained. Here the relevant margin is about how a mobile public service is deployed across space. I understand the broad conceptual similarity the authors have in mind, but I do not think the analogy is sufficiently close to motivate a distinct contribution to the public/org. econ. literature along this dimension. I would therefore suggest dropping or substantially de-emphasizing this part of the introduction.

Response to R1.5:

Thanks for the suggestion. We have removed the paragraph discussing this analogy from the introduction.

Table II: Effects of *Plan Cuadrante* Controlling for Police Resources — Column (2) Without an Upper Year Bound

	(1)	(2)	(3)	(4)	(5)
Panel A. Victimization					
ATT	-0.099***	-0.077***	-0.086**	-0.094***	-0.087*
	(0.013)	(0.013)	(0.043)	(0.028)	(0.053)
N	93,041	53,973	53,553	72,017	53,553
Panel B. Crime perception					
ATT	-0.149***	-0.140***	-0.112**	-0.082**	-0.111
	(0.025)	(0.032)	(0.057)	(0.033)	(0.106)
N	89,578	52,160	51,746	69,434	51,746
Panel C. Investment in security measures					
ATT	-0.077***	-0.066***	-0.109***	-0.081***	-0.168***
	(0.016)	(0.019)	(0.041)	(0.019)	(0.064)
N	93,041	53,973	53,553	72,017	53,553
Panel D. Reported crime (per 100k pop.)					
ATT	-208.024***	-157.010***	29.230	-261.439	-245.555
	(56.130)	(56.224)	(147.261)	(170.954)	(165.221)
N	4,231	2,963	1,911	2,248	1,911
Sample	Full	2006 onward	Resource yrs	Resource yrs	Resource yrs
Police officers (per 100k pop.)			✓		✓
Patrol vehicles (per 100k pop.)				✓	✓

Note: ATT estimates of the effect of *Plan Cuadrante* on each outcome. Column (1) is the baseline (full estimation sample, no controls) and reproduces the headline estimate of the corresponding estimator. Column (2) includes no control variable and restricts the estimation to the years from 2006 onward without imposing an upper bound; in the administrative panel it therefore extends to 2016, beyond the period covered by the police-resource data. Columns (3)–(5) add time-varying controls for police resources per 100,000 inhabitants: officers only; patrol vehicles only; and officers and patrol vehicles jointly (entered as separate regressors). Columns (3) and (5) cover 2006–2012, the years for which the number of officers is observed; column (4) also includes 2005, the first year for which patrol vehicles are observed. Patrol vehicles are measured, per 100,000 inhabitants, as all patrol automobiles (radiopatrulla, auto patrullero, auto policial, automovil, station wagon, jeep and camioneta) plus motorcycles. Standard errors in parentheses; row N reports the estimation-sample size. Estimation follows **?**; standard errors clustered at the municipality level using wild bootstrapping procedures. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

References

Referee 2 Suggestions, Comments and Questions

Thank you very much for your careful reading of the revised manuscript and response, and for confirming that it reads much better as a result. We have carefully considered all your remaining comments, in particular your suggestions on the presentation of the spillover evidence, and hope that we adequately addressed them in the revised manuscript and in the response document below.

Comment R2.1

Regarding my main concern—spillovers—the current version is considerably more transparent, and Section 4.2, together with the Online Appendix, attempts to address this threat to identification. I would, however, encourage the authors to reorganize the presentation of this evidence. My suggestion would be to lead with the exposure-based specification, in which exposure to treated neighbors is measured continuously. In my view, this is the most informative of the three exercises because it exploits variation in the intensity of potential spillovers rather than relying on a binary indicator for having a treated neighbor. This specification should, of course, be presented with the appropriate caveat: exposure to treated neighbors is not randomly assigned, so the estimates do not fully address the possibility that municipalities with many early-adopting neighbors differ systematically in other respects.

Response to R2.1:

Thanks. We have reorganized the presentation of the evidence on spillovers, as suggested. Section 4.2 now leads with the exposure-based specification following Crépon et al. (2013), which measures exposure to treated neighbors continuously (as the share and the number of treated neighbors), and only then presents the binary neighbor-treatment specification. As you note, the exposure-based specification is the more informative of the two, since it exploits variation in the intensity of potential spillovers rather than relying on a binary indicator for having a treated neighbor.

When commenting on the results of the exposure-based specification, we have introduced the recommended caveat, which now reads as follows in the paper: "Because exposure to treated neighbors is not randomly assigned, however, this evidence does not fully rule out the possibility that municipalities with many early-adopting neighbors differ systematically from those with fewer such neighbors in other respects that are also correlated with our outcomes."

Separately, and to keep the spillovers discussion focused, we have also removed the randomization-inference exercise from this section, since (as discussed in our response to your next comment) it is better understood as a general robustness check on our main estimates than as a targeted test for spillovers. We now report it separately when discussing our main results, with the full exercise moved to its own subsection in the Online Appendix (Section C.7).

Comment R2.2

I also had difficulty following the randomization-inference exercise for spillovers. This may simply reflect my own misunderstanding, but although I understand randomization inference in general, the purpose of this particular exercise as a test for spillovers was unclear to me. It may therefore warrant further explanation. In particular, how does the exercise help address concerns about spillovers? Why is treatment status reshuffled across all 346 municipalities when significantly fewer municipalities were ever eligible for the program?

Response to R2.2:

Thanks for pointing this out, and for pushing us to explain the purpose of this exercise more clearly.

This test is inspired by Blattman et al. (2021), who study a hot-spot policing experiment in which a street segment's treatment and spillover status depend on its distance to a randomly treated segment (within 250m, within 500m, and so on). Because spillover status is assigned by spatial proximity rather than by a clean, observable partition of the sample, nearby segments end up sharing the same experimental condition purely by chance. This generates what the authors call "fuzzy clustering": implicit clusters that do not correspond to any observed grouping variable, that vary in size, and whose size can be correlated with potential outcomes. Standard cluster-robust standard errors assume a well-defined partition and can be biased in this setting—not only the standard errors, but potentially the point estimates themselves. Blattman et al. (2021) address this with randomization inference: they repeatedly re-simulate their own (staggered, two-stage) randomization procedure to obtain an exact reference distribution that remains valid despite this clustering structure.

Our setting differs in two important respects. First, and most importantly, treatment in our paper—the adoption of *Plan Cuadrante*—is not randomly assigned, unlike the experiment in Blattman et al. (2021). Second, our treatment and spillover status are assigned at the municipality level (territorial units that often exceeds 2000 sq km), which is also our clustering unit for inference; municipality boundaries, unlike distance-based spillover bands, give us a clean partition of the sample, so we do not face an analogous "fuzzy clustering" problem. Taken together, we do not believe our standard errors suffer from the bias that motivates randomization inference in Blattman et al. (2021).

We agree with the editors that, for these same reasons, this exercise is not as informative about spillovers in our setting as it is in Blattman et al. (2021): absent a true random-assignment mechanism, permuting adoption dates and never-treated status does not yield the same kind of valid, design-based test for spillovers that randomization inference provides in their experimental setting. We agree with your characterization: in our case the exercise is better understood as a general robustness/placebo check on our staggered difference-in-differences design than as a targeted spillover test or a correction to our standard errors. In that spirit, we keep the exercise but now report it in its own subsection of the Online Appendix (Section C.7), with a brief mention when discussing our main results (Section 4.1) rather than in the spillovers discussion, and no longer present it as direct evidence on spillovers. We also revise the exercise itself in response to your specific question: rather than reshuffling adoption dates and never-treated status across the full national sample of 346 municipalities—most of which were below the 70% urban-population threshold and were therefore never eligible for *Plan Cuadrante*, and so could not have adopted it under any circumstance—we now restrict the placebo exercise to the municipalities that eventually adopted *Plan Cuadrante*, randomly reassigning each municipality's own adoption date within the period for which we observe it. This keeps the placebo exercise anchored to municipalities we know were capable of receiving the treatment, instead of mixing in municipalities that were never at risk of adoption. One consequence of this change is that the placebo sample no longer contains any genuinely untreated municipalities, so we can no longer construct an analogous randomization-inference test for spillovers; we have therefore dropped that comparison from Online Appendix Figure C10 and no longer report a spillover result from this exercise. Spillovers continue to be addressed through the exposure-based and binary neighbor-treatment specifications discussed in Section 4.2.

Comment R2.3

I found a few small inconsistencies in the response to my report—for example, references to Figure 4 where Figure 6 appears to have been intended—but I checked these against the revised manuscript and did not find the same inconsistencies there. There is one exception. In response to one of my comments, the authors quote revised text for Online Appendix C.1 stating that "the effects on arrests are not statistically significant in the ENUSC subsample." This sentence does not appear in the manuscript, which still reads only: "While the effects for reported crime are larger in magnitude, the estimates are qualitatively similar..."

Response to R2.3:

Thanks for catching this, and apologies for the inconsistency. Regarding the remaining inconsistency in the manuscript: That sentence was lost during the last round of edits among co-authors; it was never a discrepancy in the underlying results, which have consistently shown a statistically insignificant arrests estimate in the ENUSC subsample (the table reports an ATT of 44.09 with no significance stars in that subsample, versus 48.02, significant at the 1% level, in the full administrative sample). We have restored the missing sentence in Online Appendix C.1, which now reads: "The estimated effect on arrests, however, is not statistically significant in this smaller ENUSC subsample, likely reflecting reduced power given the smaller number of municipalities and observations."

Comment R2.4

I also noticed that some estimates based on the administrative data have changed between the two versions. In particular, the estimated reduction in reported crime is now 10 percent rather than 14 percent. I could not find this change documented in the response. As long as the authors can provide a reasonable explanation for what changed between rounds and are confident that the current result is robust, I do not view this as a major concern.

Response to R2.4:

Thanks for pointing this out, and apologies for not documenting this change explicitly in our previous response.

The change in the headline number—from a 14% to a 10% decline in reported crime— comes mainly from a change in how we compute $\hat{Y}$, the baseline level we use as the denominator to convert that level effect into a percentage change. In the previous version, $\hat{Y}$ was computed as a simple average over the pooled sample of observations—pre-treatment observations of treated municipalities together with observations from never-treated municipalities—rather than as an average of the two groups' separate means. Since the Callaway and Sant'Anna estimator is essentially a treatment-on-the-treated (ToT) parameter, we now believe it is more appropriate to compute $\hat{Y}$ using only the pre-treatment observations of treated municipalities. This revision affects the administrative-data-based estimates more evidently than the ENUSC-based ones, because the administrative sample includes 196 never-treated municipalities, whereas the ENUSC sample includes only one; pooling into $\hat{Y}$ therefore moved the administrative baseline by much more than the ENUSC one. We apologize again for not flagging this change explicitly in our earlier response. To avoid this ambiguity going forward, we now define $\hat{Y}$ explicitly in the notes of every figure that reports it.

A second, unrelated and more minor change may also explain very minor differences in some β coefficients (including the headline estimate). Between the first and second submission we realized that our implementation of the Callaway and Sant'Anna estimator was not weighting estimates by cohort size, while the default "simple" and "dynamic" aggregation schemes in Callaway and Sant'Anna (2021) weight by cohort size (DANI INCLUYE AQUI ALGUNA CITA PORFA – adicional a Callaway, si tienes una referencia mas especifica sobre el uso de este peso por defecto en la practica, mejor). We corrected this, which slightly changes some point estimates but not in a way that is economically or statistically meaningful.

Comment R2.5

Finally, I noted a few minor editorial issues:

1. Two spaces are missing: on p. 7, "full municipal territory.On average," and on p. 21, "improve security perceptions,police forces need."

Response to R2.5:

Thanks for the catch. Both missing spaces are now corrected in the updated manuscript.

Comment R2.6

2. The abbreviation "PCSP" does not appear to be defined in the paper, although it is used in some notes and figures.

Response to R2.6:

Thanks for the catch. We now introduce the program's full name, Plan Cuadrante de Seguridad Pública, the first time it is mentioned in the institutional background section, noting that for simplicity we refer to it as Plan Cuadrante throughout. We also replaced the remaining instances of "PCSP" in the paper with "PC" for consistency.

Comment R2.7

3. The reference list includes Ajzenman, Dominguez, and Undurraga (2023a) and (2023b) as separate entries for what appears to be the same paper, and both entries are cited in the text.

Response to R2.7:

Thanks for the catch. This is now corrected: we removed the duplicate entry and all in-text citations now point to the single remaining reference.

References

Blattman, C., D. P. Green, D. Ortega, and S. Tobón (2021, 02). Place-Based Interventions at Scale: The Direct and Spillover Effects of Policing and City Services on Crime. *Journal of the European Economic Association 19*(4), 2022–2051.

Callaway, B. and P. H. Sant'Anna (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics 225*(2), 200–230.

Crépon, B., E. Duflo, M. Gurgand, R. Rathelot, and P. Zamora (2013). Do labor market policies have displacement effects? Evidence from a clustered randomized experiment. *The Quarterly Journal of Economics 128*(2), 531–580.